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JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY **8875/02**

Paper 2 **2 September 2013**

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.
Circle the question number of the question attempted.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
5 / 6	
Total	

This document consists of **12** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows a cell that has undergone semi-conservative replication and is in the process of undergoing telophase and cytokinesis.

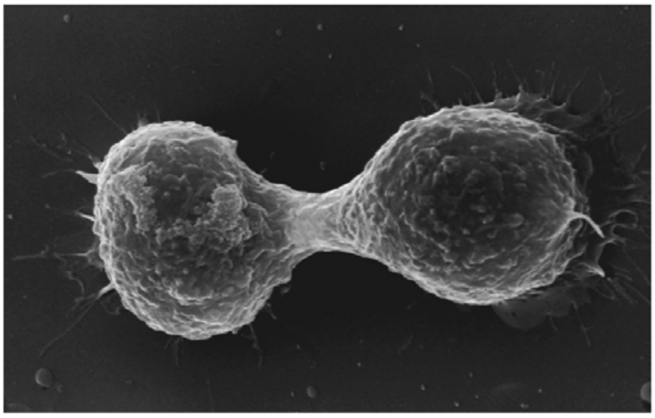

Fig. 1.1

(a) (i) State precisely when DNA replication occurs in the cell cycle. [1]

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(ii) Outline the main features of semi-conservative DNA replication. [5]

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(b) Describe the events that occur during the stage shown in Fig. 1.1. [3]

(c) State how mitosis maintains genetic stability in an organism. [1]

[Total: 10]

2 Fig. 2.1 is a diagram of an electron micrograph of part of a chloroplast showing the thylakoid membranes.

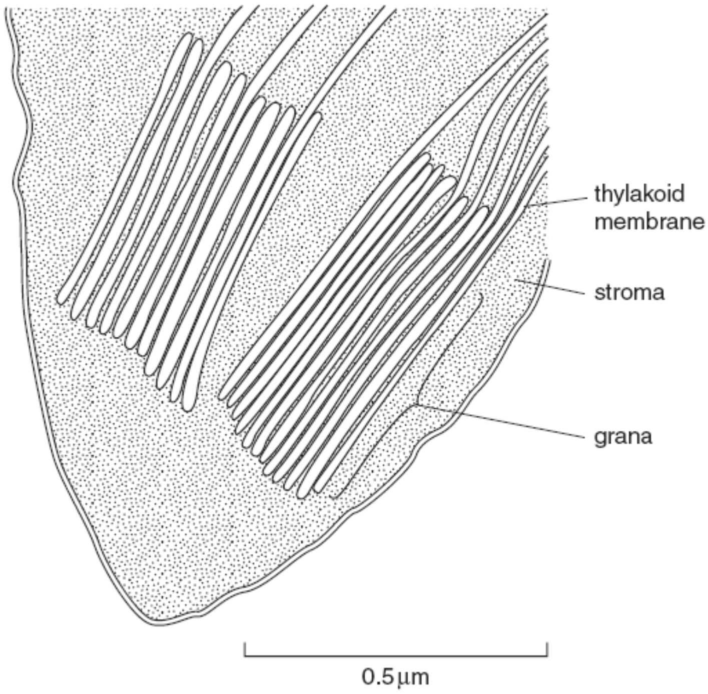

Fig. 2.1

(a) Describe the role of the thylakoid membrane in photosynthesis. [3]

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(b) List three ways in which photophosphorylation differs from oxidative phosphorylation. [3]

(c) Fig. 2.2 shows the results from two experiments carried out to investigate the effect of light intensity and carbon dioxide concentration on the rate of photosynthesis.

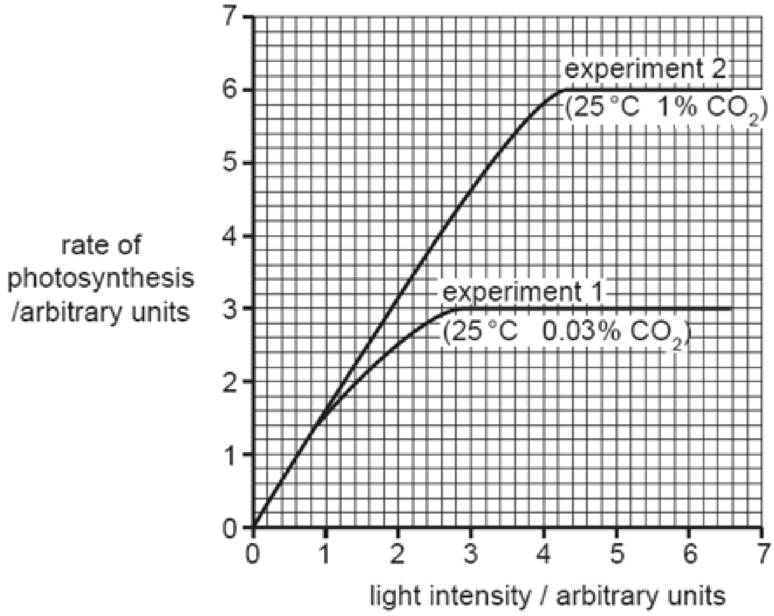

Fig. 2.2

(i) Describe and explain the results shown in Fig. 2.2 for experiment 1. [3]

To further investigate the effects of carbon dioxide concentration on the Calvin cycle, the levels of ribulose bisphosphate (RuBP) and glycerate-3-phosphate (GP) were determined and the results shown in Fig. 2.3.

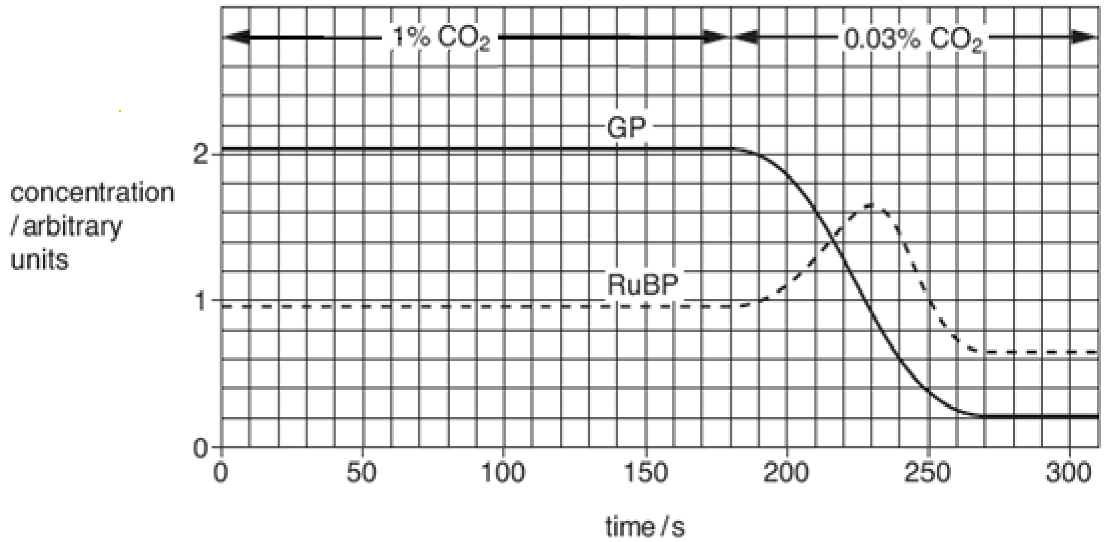

Fig. 2.3

(ii) Explain why the concentration of RuBP increases during the 50 seconds immediately after the carbon dioxide concentration is reduced to 0.03%. [2]

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[Total: 11]

3 The takahē, *Porphyrio hochstetteri*, is a flightless bird that is restricted to a small area of the South Island of New Zealand. It is one of only two remaining species of large, flightless, herbivorous birds from New Zealand. The other remaining species is found on the North Island of New Zealand. Their flighted ancestor came over from Australia millions of years ago.

The takahē was thought to be extinct, but a small population with low genetic variation was discovered in 1948 among the remote mountains of the South Island. The takahē is a grassland specialist and lives in alpine grassland dominated by tussock grasses as shown in Fig. 3.1.

Fig. 3.1

(a) Explain why it has been possible for flightless birds, such as the takahē, to evolve in New Zealand. [3]

(b) Suggest why low genetic variation in the population of takahē is undesirable. [1]

(c) To increase genetic variation in the population, conservationists selected some birds with more variation in their alleles and allowed them to interbreed. This is an example of artificial selection.

(i) Suggest one difference between artificial selection and natural selection. [1]

(ii) Describe two processes that led to increased genetic variation in the population. [2]

Fig. 3.2 shows the relationship between some animals based on their anatomical homology by comparing the arrangement of the bones in their forelimbs which had a basic pentadactyl structure.

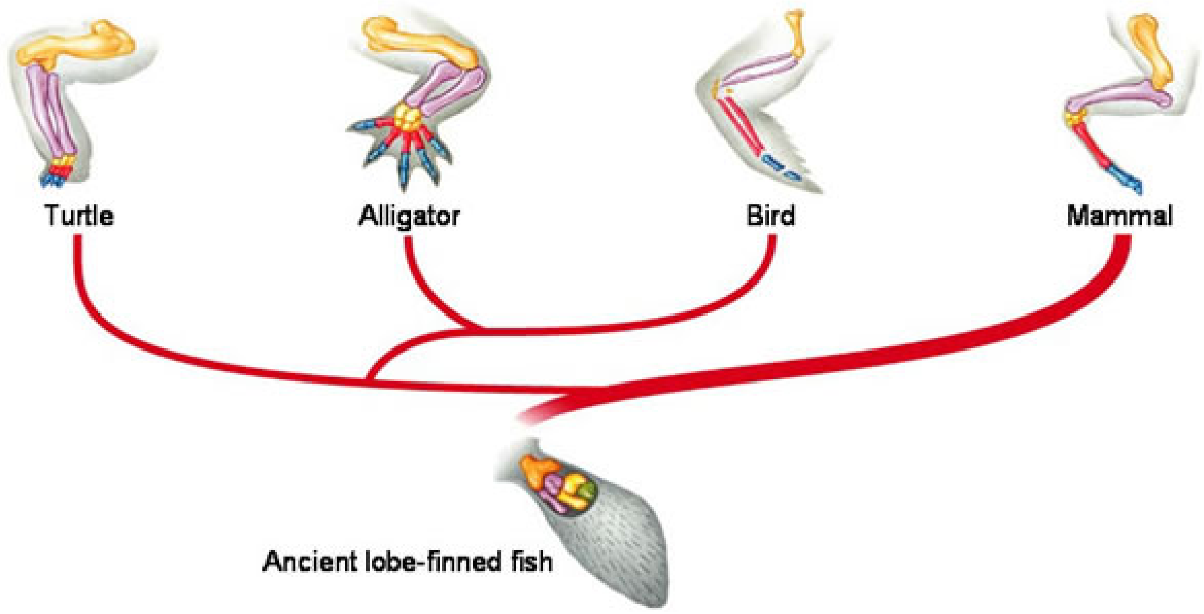

Fig. 3.2

(d) Explain how the anatomical homology in Fig. 3.2 provides evidence to support the theory of evolution. [3]

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[Total: 10]

4 Insulin deficiency is a widespread problem in developed countries. To alleviate the symptoms, insulin injection is given to these patients. Insulin is manufactured on a large scale basis using genetic engineering techniques involving restriction enzymes and plasmids.

(a) Explain what is meant by a restriction enzyme. [2]

Plasmids used in genetic engineering include pMOD2-Neo shown in Fig. 4.1. The pMOD2-Neo plasmid consists of 2871 bp and has an origin of replication (ori), a multiple cloning site (MCS) with restriction sites for 9 different restriction enzymes, a gene giving resistance to the antibiotic neomycin (neo) and a gene giving resistance to the antibiotic ampicillin (amp).

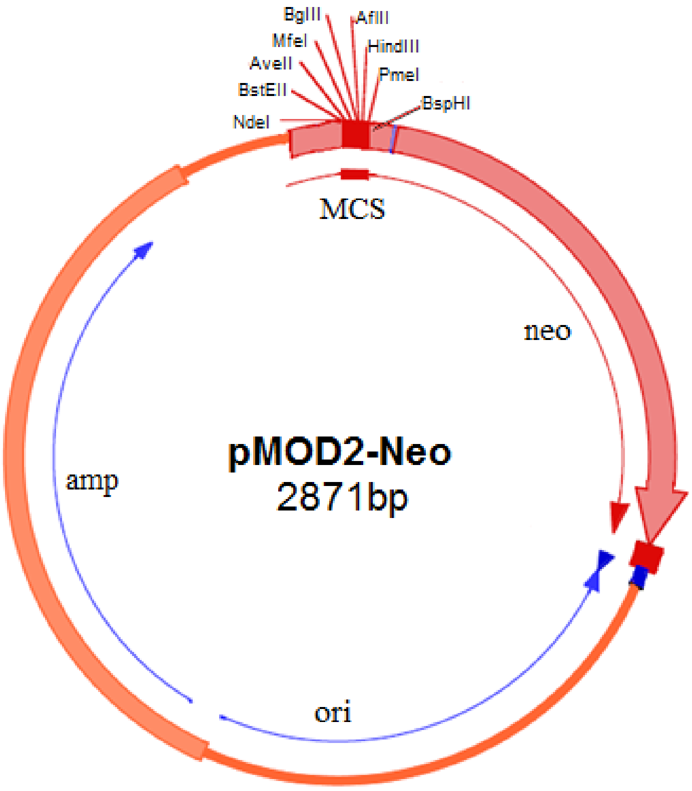

Fig. 4.1

(b) With reference to Fig. 4.1, describe how each region of the plasmid allows it to be used for cloning:

(i) Origin of replication [1]

(ii) Ampicillin resistance gene [1]

(iii) Multiple cloning site [2]

(c) Scientists have developed another treatment for these patients, such as the use of stem cells which can be stimulated to develop into pancreatic cells that can synthesise and release insulin.

State the normal functions of the stem cells used during the treatment. [2]

(d) Discuss the differences between adult stem cells and embryonic stem cells. [1]

[Total: 9]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

5 **(a)** Explain how temperature affects the rate of an enzyme-catalysed reaction. [8]

(b) Outline the ways in which mutations can occur. [6]

(c) Discuss the social implications of genetically modified organisms. [6]

[Total: 20]

Or

6 **(a)** Describe the process of glycolysis and Krebs cycle. [10]

(b) Describe how the eukaryotic gene can be incorporated into *E.coli*. [4]

(c) Discuss the benefits of the Human Genome Project. [6]

[Total: 20]